

Part 1 : Sequential Forward Selection

1. 所有按照 SFS 選取排序的特徵，以及所對應的 Accuracy

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Part1: Sequential Forward Selection (SFS)
-----Sequential Forward Selection (SFS)-
Step 1: Selected Feature 27, Balanced Accuracy: 90.49%
Step 2: Selected Feature 22, Balanced Accuracy: 92.70%
Step 3: Selected Feature 1, Balanced Accuracy: 94.39%
Step 4: Selected Feature 6, Balanced Accuracy: 94.46%
Step 5: Selected Feature 15, Balanced Accuracy: 94.62%
Step 6: Selected Feature 10, Balanced Accuracy: 94.83%
Step 7: Selected Feature 21, Balanced Accuracy: 95.19%
Step 8: Selected Feature 18, Balanced Accuracy: 95.32%
Step 9: Selected Feature 14, Balanced Accuracy: 95.53%
Step 10: Selected Feature 28, Balanced Accuracy: 95.41%
Step 11: Selected Feature 3, Balanced Accuracy: 95.72%
Step 12: Selected Feature 26, Balanced Accuracy: 96.53%
Step 13: Selected Feature 17, Balanced Accuracy: 96.94%
Step 14: Selected Feature 25, Balanced Accuracy: 96.94%
Step 15: Selected Feature 16, Balanced Accuracy: 96.63%
Step 16: Selected Feature 4, Balanced Accuracy: 97.16%
Step 17: Selected Feature 13, Balanced Accuracy: 97.05%
Step 18: Selected Feature 24, Balanced Accuracy: 97.31%
Step 19: Selected Feature 11, Balanced Accuracy: 97.32%
Step 20: Selected Feature 19, Balanced Accuracy: 97.32%
Step 21: Selected Feature 29, Balanced Accuracy: 97.10%
Step 22: Selected Feature 5, Balanced Accuracy: 96.11%
Step 23: Selected Feature 23, Balanced Accuracy: 96.53%
Step 24: Selected Feature 8, Balanced Accuracy: 96.55%
Step 25: Selected Feature 9, Balanced Accuracy: 95.59%
Step 26: Selected Feature 0, Balanced Accuracy: 93.46%
Step 27: Selected Feature 2, Balanced Accuracy: 94.04%
Step 28: Selected Feature 7, Balanced Accuracy: 91.17%
Step 29: Selected Feature 20, Balanced Accuracy: 93.45%
Step 30: Selected Feature 12, Balanced Accuracy: 93.93%
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2. 最佳的準確率及其對應的特徵子集合

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[-] Best Balanced Accuracy: 97.32 %
[-] Optimal Feature Subset (indices): [27, 22, 1, 6, 15, 10, 21, 18, 14, 28, 3, 26, 17, 25, 16, 4, 13, 24, 11]
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Part 2 Fisher's Criterion

1. 按照 Fisher's Criterion 選取排序的特徵，及其對應的 Accuracy

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Part2: Fisher's Criterion
-----Fisher's Criterion-----
Fisher's scores (sorted): [3.39164828e+00 2.81287360e+00 2.70300272e+00 2.69992209e+00
2.25356327e+00 2.10359068e+00 1.93957084e+00 1.75113798e+00
1.73278631e+00 1.53596316e+00 1.04556630e+00 9.83969043e-01
8.01869788e-01 7.50703185e-01 6.91208134e-01 5.63899010e-01
4.50222345e-01 4.49965701e-01 4.29469485e-01 3.86939706e-01
3.19615346e-01 2.54161481e-01 2.23052361e-01 1.94180874e-01
1.61581388e-01 1.42004753e-02 9.75605413e-03 3.40988820e-04
1.54316992e-04 8.25866505e-05]
Top-1 features: [27], Balanced Accuracy: 90.49%
Top-2 features: [27 22], Balanced Accuracy: 92.70%
Top-3 features: [27 22 7], Balanced Accuracy: 92.18%
Top-4 features: [27 22 7 20], Balanced Accuracy: 92.38%
Top-5 features: [27 22 7 20 2], Balanced Accuracy: 90.41%
Top-6 features: [27 22 7 20 2 0], Balanced Accuracy: 90.89%
Top-7 features: [27 22 7 20 2 0 23], Balanced Accuracy: 93.64%
Top-8 features: [27 22 7 20 2 0 23 6], Balanced Accuracy: 76.09%
Top-9 features: [27 22 7 20 2 0 23 6 3], Balanced Accuracy: 92.54%
Top-10 features: [27 22 7 20 2 0 23 6 3 26], Balanced Accuracy: 92.89%
Top-11 features: [27 22 7 20 2 0 23 6 3 26 5], Balanced Accuracy: 91.96%
Top-12 features: [27 22 7 20 2 0 23 6 3 26 5 25], Balanced Accuracy: 91.36%
Top-13 features: [27 22 7 20 2 0 23 6 3 26 5 25 10], Balanced Accuracy: 91.36%
Top-14 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12], Balanced Accuracy: 90.79%
Top-15 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13], Balanced Accuracy: 92.55%
Top-16 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21], Balanced Accuracy: 95.51%
Top-17 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1], Balanced Accuracy: 95.11%
Top-18 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24], Balanced Accuracy: 92.51%
Top-19 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17], Balanced Accuracy: 92.07%
Top-20 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28], Balanced Accuracy: 91.53%
Top-21 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4], Balanced Accuracy: 89.43%
Top-22 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8], Balanced Accuracy: 92.23%
Top-23 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29], Balanced Accuracy: 90.72%
Top-24 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15], Balanced Accuracy: 92.94%
Top-25 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16], Balanced Accuracy: 78.85%
Top-23 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29], Balanced Accuracy: 90.72%
Top-24 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15], Balanced Accuracy: 92.94%
Top-25 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16 19], Balanced Accuracy: 79.24%
Top-27 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16 19 14], Balanced Accuracy: 86.91%
Top-28 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16 19 14 9], Balanced Accuracy: 91.40%
Top-29 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16 19 14 9 11], Balanced Accuracy: 91.92%
Top-30 features: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21 1 24 17 28 4 8 29 15
16 19 14 9 11 18], Balanced Accuracy: 93.93%
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2. 最佳的準確率及其對應的特徵子集合。

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[-] Best Balanced Accuracy: 95.51 %
[-] Optimal Feature Subset (indices): [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21]
[-] Number of Features: 16
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Part 3 Discussion and results presenting

比較結果：

SFS 最佳 Balanced Accuracy: 97.32%

SFS 最佳特徵子集合: [27, 22, 1, 6, 15, 10, 21, 18, 14, 28, 3, 26, 17, 25, 16, 4, 13, 24, 11]

Fisher' s Criterion 最佳 Balanced Accuracy: 95.51%

Fisher' s Criterion 最佳特徵子集合: [27 22 7 20 2 0 23 6 3 26 5 25 10 12 13 21]

1. Sequential Forward Selection 和 Fisher's Criterion 分別屬於 Filter-based 和 Wrapper-based 中的何種特徵篩選方法？

Sequential Forward Selection (SFS) 是 Wrapper-based 特徵篩選方法，因為它使用分類器的效能來評估特徵子集合。

Fisher' s Criterion 則是 Filter-based 特徵篩選方法，因為它基於統計量來評估特徵的重要性，不依賴於特定的分類器。

2. 一般來說 Filter-based 和 Wrapper-based 各有什麼性質或優缺點？

Filter-based 方法通常計算速度較快，且不依賴於特定的分類器，但可能忽略特徵間的交互作用。Wrapper-based 方法則能考慮特徵間的交互作用，通常能獲得更好的分類效能，但計算成本較高。

3. 在本次作業的結果中是否有展現出跟上一題你的回答有一致的現象呢? 不管是否一致皆請你試著討論與分析原因。

在本次作業的結果中，SFS 和 Fisher' s Criterion 都能有效地選擇出具有區分性的特徵子集合。具體哪種方法表現更好，取決於資料集的特性和特徵之間的關係。這與 Filter-based 和 Wrapper-based 方法的性質一致。